HealthConnect Rwanda

Healthcare Operations and Performance Analytics

Executive presentation | Data Analytics Track | Week 6

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RWF 919.9M is uncollected, and no single hospital stands out as the problem

20,000
patients, 60,000 billed visits

90.1%
collected: RWF 8.39bn of 9.31bn

74.6%
of patients have a chronic condition

2.99 / 5
average satisfaction; 40.4% poor

- Money: 19.8% of bills are short-paid, and 4,042 bills marked "Paid" still carry RWF 311.2M.
- Claims: 50% of insurance claims (RWF 2.44bn) are pending and none is rejected.
- Patterns: staffing, diagnoses and provinces are nearly uniform, so the issues are system-wide, not one site.

From 27 datasets to one validated database and a six-page dashboard

1 Load

31 tables into PostgreSQL 18 with exact column types

2 Validate

44 of 44 row-count and total checks pass

3 Analyse

40 SQL questions, each recomputed in Python

4 Visualise

Power BI: 6 pages, 30+ DAX measures

Differences are reported as findings only when they exceed random variation (chi-square and Kruskal-Wallis tests).

Four data gaps shape what the numbers can claim

Appointments: 5,000 of 60,000

91.7% of bills cannot be linked to a facility, department, doctor or date.

No facility master

Facilities are IDs only (HF001 to HF020): no names, type or location.

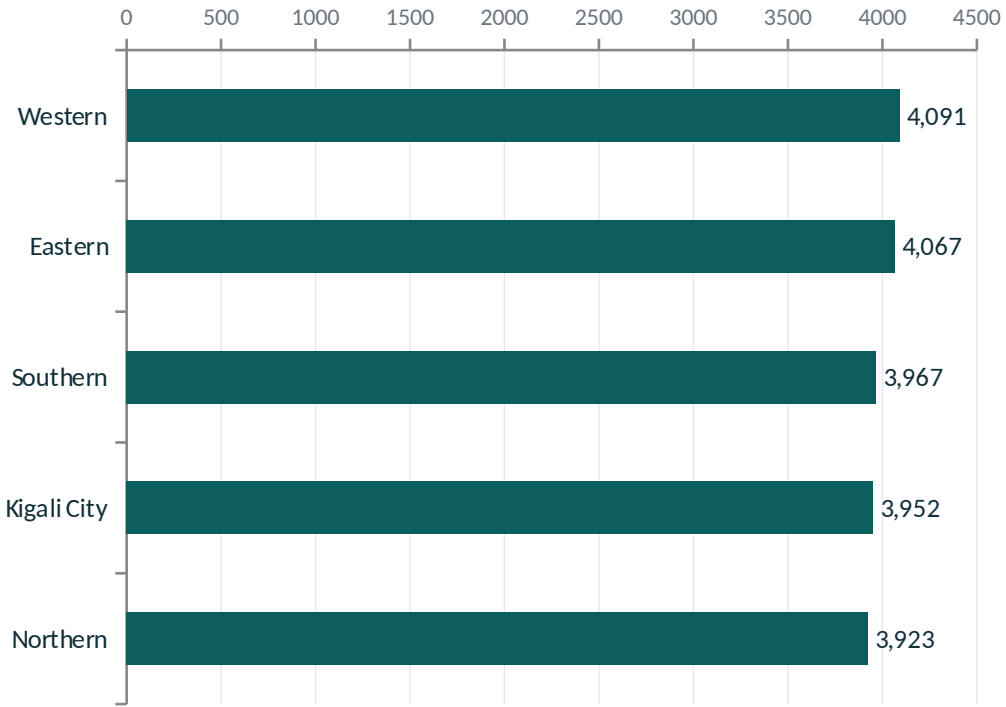
Revenue does not reconcile

Billing RWF 9.31bn, payments 8.39bn, revenue summary only 1.13bn.

Missing outcomes

No rejected claims, bed occupancy, equipment status or cost data.

Patients are spread evenly across provinces, and three in four are chronic



74.6%

have a chronic condition (Asthma, Hypertension, Diabetes at about 25% each)

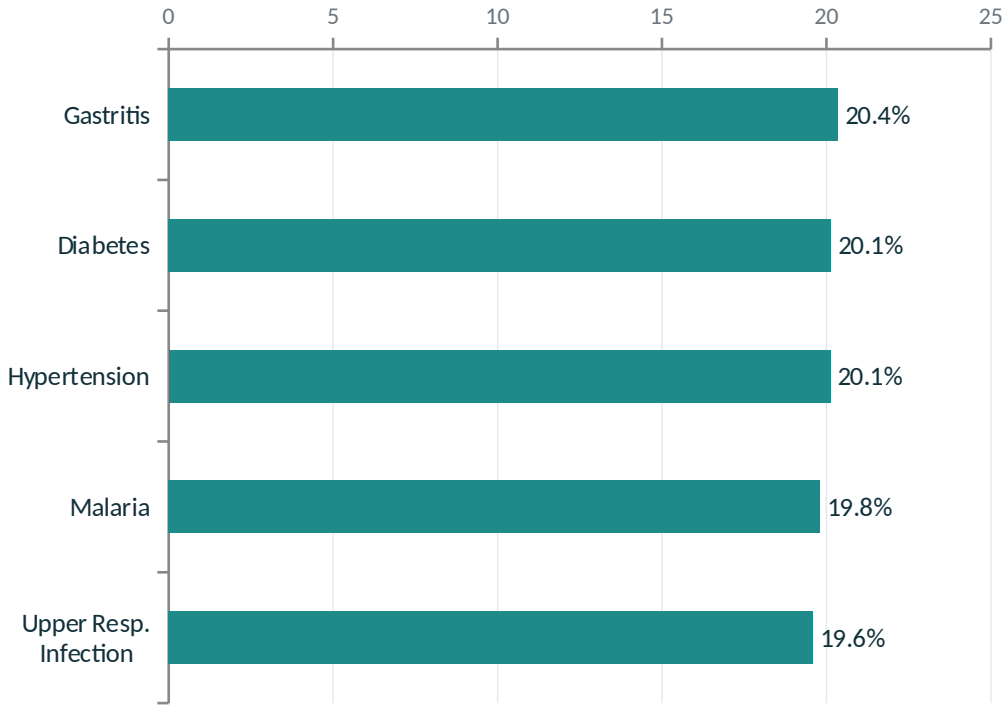
27.4%

are over 65 (5,487 patients)

45.0

average age; 50.5% male

The clinical mix is even: five diagnoses at about 20% each, and no trend



33.3%
severe cases

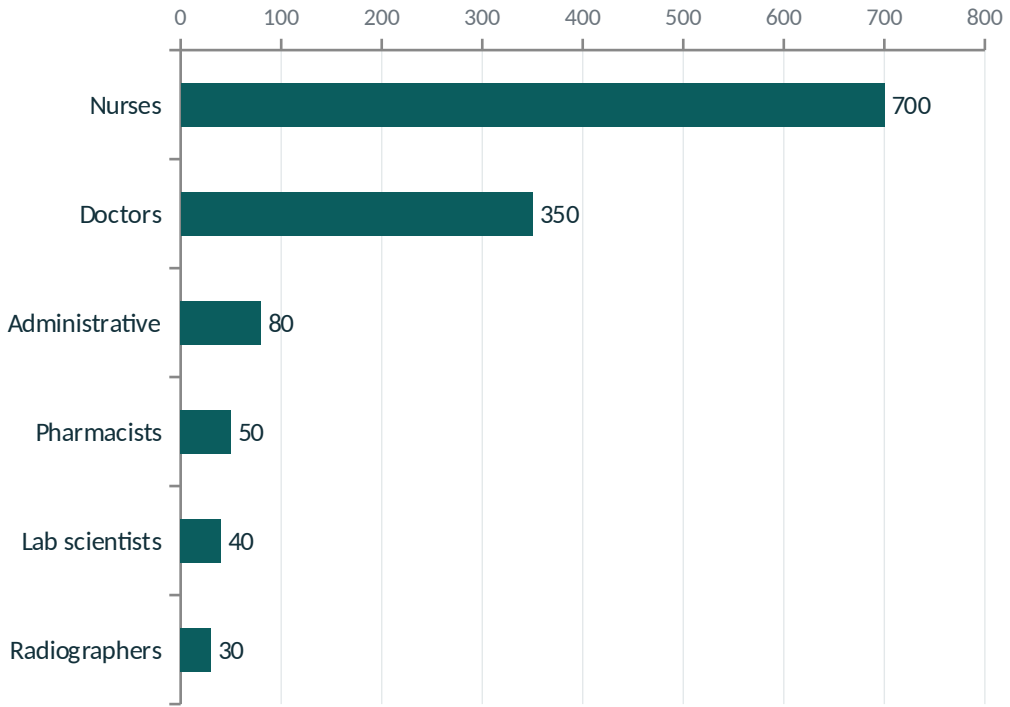
8.3%
admission rate

5.49
days average stay

7.85%
referral rate

Plan services for all five conditions in similar proportion. Dates exist for only 5,000 diagnoses, so trends need fuller data.

Staffing is uniform, but demand per facility varies by about 30%



1,250
staff, 20 facilities

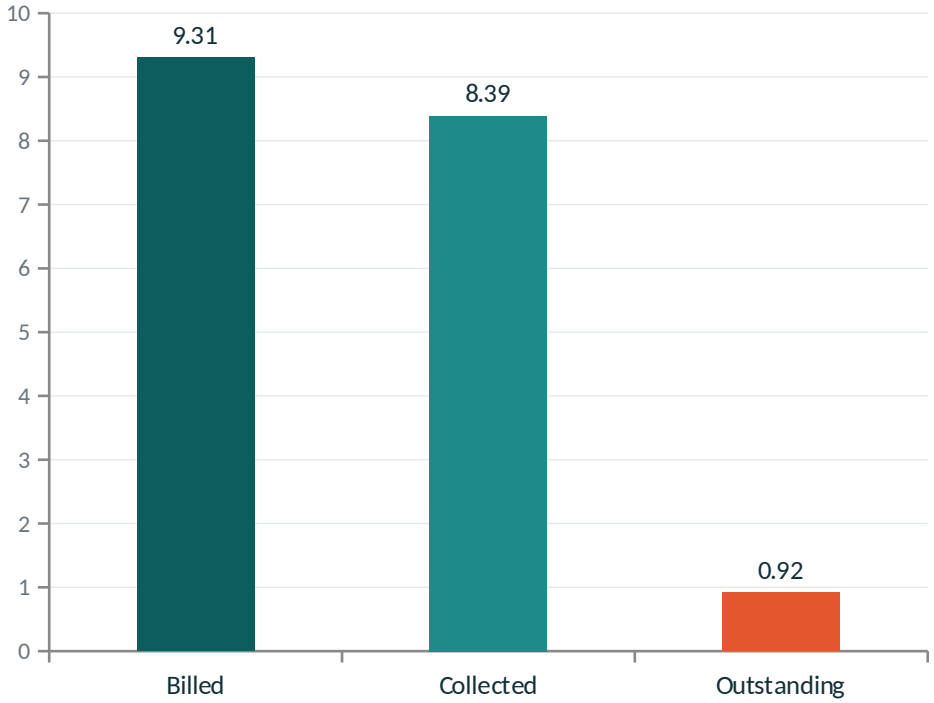
62.5
staff per 1,000 patients

33.0%
of visits are emergency

3.8%
estimated bed use

2,000 beds are all marked Available. Bed use is estimated from admissions (27,438 bed-days of 730,000), so expansion cannot be justified yet.

RWF 919.9M is outstanding, and "Paid" bills still carry RWF 311.2M



11,887

bills (19.8%) paid short; average shortfall RWF 77,391

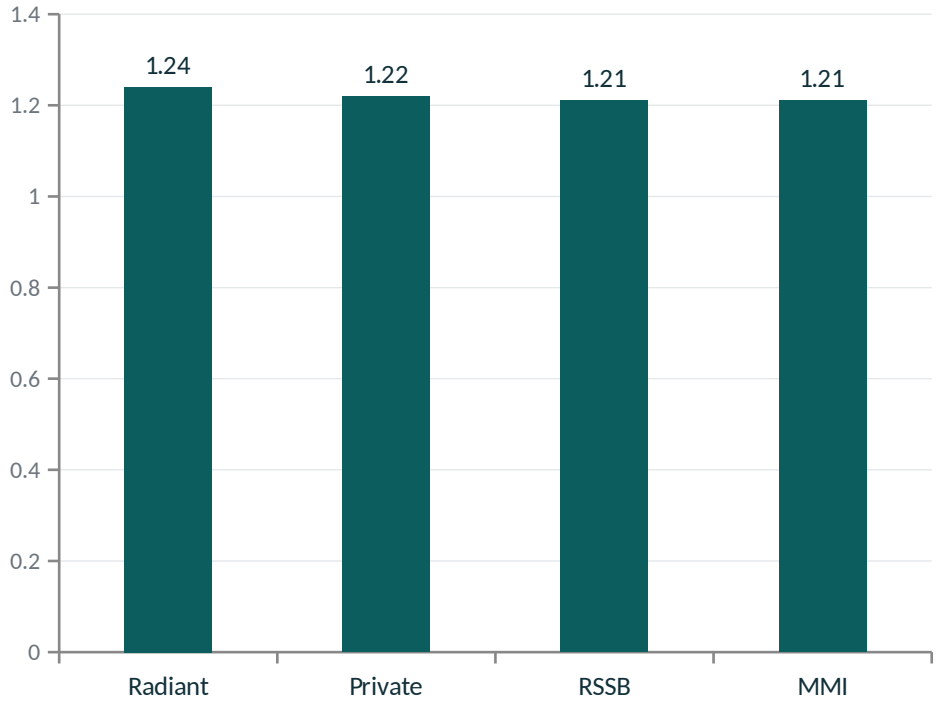
4,042

bills marked "Paid" still have balances (RWF 311.2M)

25%

of value per payment method: card, cash, insurance, mobile money

Half of insurance claims are pending, and none are rejected



70.1%

of bills have an insurance claim

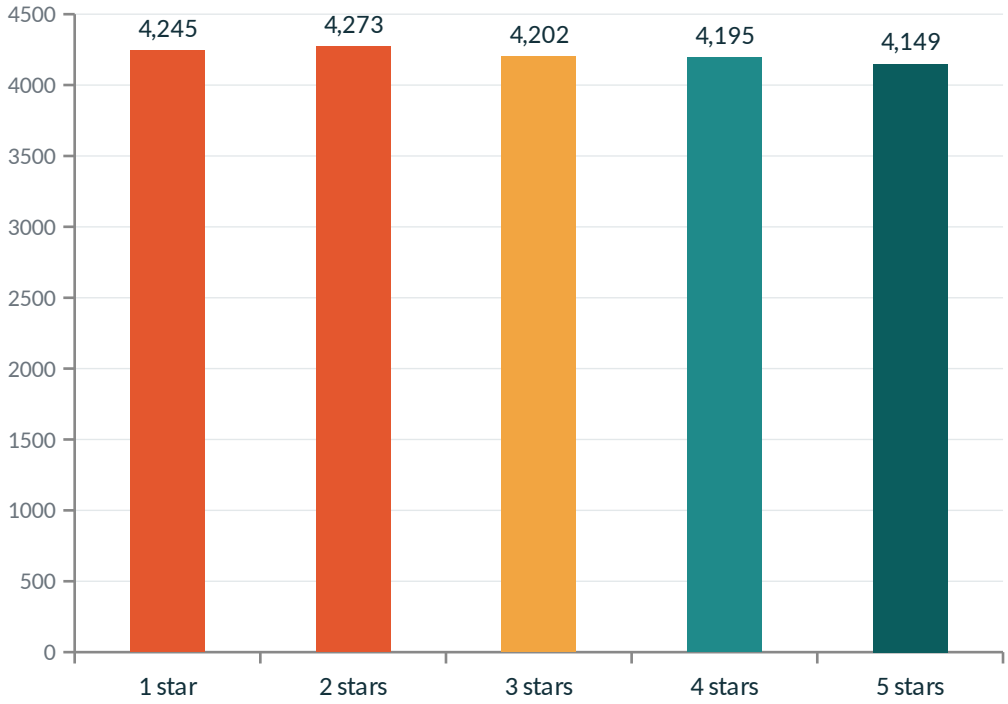
RWF 2.44bn

in pending claims (50.0%)

0

claims marked Rejected: tracking gap

Satisfaction is low everywhere: 40% of ratings are poor, 20% excellent



2.99 / 5

average across 21,064 ratings

40.4% vs 19.7%

poor (1 to 2) versus excellent (5)

Only 1,756 ratings link to a facility. Feedback comments do not match star ratings, so ratings are used.

Answers to the ten business questions

Question	Answer	Confidence
Most patients	HF013, HF014, HF010 (sample); small spread	Low
Greatest dept workload	Cardiology 544, Radiology 525; not significant	Low
Diseases rising	None: five at about 20%, no monthly trend	Medium
Staff efficient?	Uniform staff, demand varies 30%: review	Medium
Top insurer revenue	Radiant RWF 1.24bn; all within 2%	High
Highest satisfaction	HF011 (3.19); not significant	Low
Needs expansion	No evidence: bed use about 3.8%	Low
Cut costs	No cost data; recover RWF 919.9M, clear claims	Medium
Services by revenue	Visit types equal; Cardiology top dept (sample)	Low
Regions to invest in	None stands out; chronic 73.5% to 76.2%	Medium

Six priority actions for management

1

Recover cash

Follow up 11,887 short-paid bills; compute balances from amounts; block "Paid" when under-paid.

2

Clear claims

Resolve 21,028 pending claims; add Rejected status; reconcile with insurer payments.

3

Chronic care

Start a chronic-disease and elderly-care programme in every province.

4

Match staff to demand

Review staffing against full appointment volumes; emergency cover first.

5

Lift service quality

System-wide programme; record the reason for every low rating.

6

Fix the data

Full appointment tables, facility master, bed occupancy, claim rejections, costs.

What this analysis can and cannot show, and next steps

Can show

- Collection gaps, claim backlog and status errors in billing
- Patient profile, chronic burden and ageing
- Whether differences between sites are real or random

Cannot show

- Facility or region performance for the 91.7% of bills with no appointment
- Causes, costs or true bed occupancy
- Disease trends over time

Next steps: load the full appointment tables and an official facility master, refresh the dashboard, and re-test the findings. Thank you.